# The Hidden Variable in Embedded VLM Perception: Temporal Calibration for Real-Time Work-Zone State Estimation

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***Abstract*—Vision–language models (VLMs) are increasingly proposed to replace detector pipelines in driving perception, usually against the baseline's published temporal smoothing constants. We show on embedded hardware that this hidden variable can decide the outcome. We run an INT4 2B VLM fine-tuned on ∼46K work-zone question–answer pairs and ∼10K hard negatives and a zero-shot 4B physical-AI reasoner on a Jetson AGX Orin through an evidence-first prompt cascade, and compare them with a YOLO+CLIP pipeline under a latency-honest protocol in which measured latency drives frame dropping. On 484K annotated frames of ROADWork driving, constants inherited from the 2B leave the 4B model at macro-F1 0.290; recalibrated, it reaches 0.515. Recalibration lifts the detector baseline from 0.470 to 0.551, the best result, and no VLM-based system then beats it; a counted hidden-Markov filter on the detector score alone is statistically tied with it without a threshold search. These results hold on drive-disjoint folds and at matched false-alarm rates and search budgets, and live concurrent execution reproduces the replay; on a DRIVE AGX Thor, cycle-counted constants degrade while the counted filter holds.**

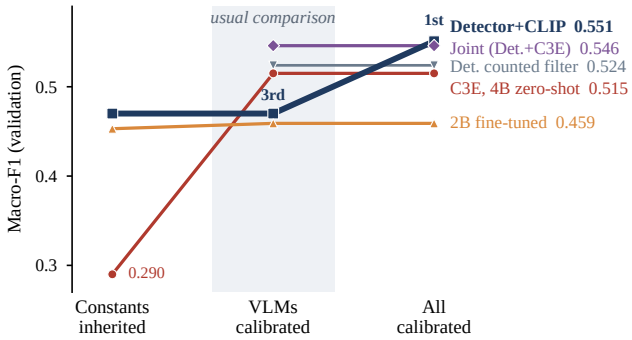



Fig. 1. The hidden variable. Validation macro-F1 (208 videos) under three comparison regimes. Left: every VLM runs the 2B's temporal constants and the detector pipeline its published ones; the zero-shot 4B reasoner (C3E) looks useless. Middle: the VLM systems are calibrated and the baseline is not, as in a typical comparison; C3E and the joint estimator beat the detector pipeline. Right: the baseline is calibrated by the same procedure and becomes the best system. Grey: the detector-only counted filter, which carries most of the joint estimator's lead.

## I. Introduction

Work zones concentrate difficult driving-perception conditions: their appearance is regional and unstandardized, geometry changes daily, evidence such as cones, barrels, signs, and workers is small or occluded, and both misses and nuisance alerts are costly. They accounted for roughly 101,000 U.S. crashes and 899 fatalities in 2023 [1], yet production ADAS rarely model work zones explicitly and temporary speed limits are often absent from digital maps.

The strongest onboard approach combines a YOLO-style work-zone detector, CLIP verification [2], exponential smoothing, and hysteresis to estimate four traversal states [3], [4]. It is fast and effective after a zone is established, but reported IoU falls from 0.56/0.47 on OUTSIDE/INSIDE to 0.11/0.09 on APPROACHING/EXITING [3]. These boundaries are precisely where an advisory is most useful: an alert after the vehicle reaches the barrels has little anticipatory value.

Meanwhile, fine-tuned VLMs describe work-zone scenes well offline [5], and weight-only quantization plus embedded runtimes bring small VLMs to automotive hardware [6], [7]. Whether they should replace detector pipelines is usually judged by comparing systems as published, each with the smoothing and hysteresis constants its authors tuned. We build the strongest embedded VLM estimator we can, run it beside the detector pipeline on the same device, videos and protocol, and ask what such a comparison actually measures.

The answer is: mostly the temporal constants (Fig. 1). Against the hand-tuned detector pipeline our VLM-based systems look competitive or better, but constants inherited from one checkpoint leave a 4B reasoner at INSIDE IoU 0.007, and its own calibration lifts it to 0.574. The same calibration raises the detector pipeline from 0.470 to 0.551, after which no VLM-based system beats it. A team choosing a perception stack from the usual comparison would adopt a costlier VLM stack for no significant gain in macro-F1.

Our contributions are three. **(1) Evaluation:** a latency-honest protocol, confirmed by live concurrent execution, with per-model recalibration by evidence replay. **(2) Findings:** temporal constants transfer neither between the systems we test nor to faster hardware, and calibrating the baseline reverses the ranking; this holds on drive-disjoint folds, at matched false alarms and with matched search budgets. **(3) Systems:** a real-time evidence-first cascade for a quantized 2B VLM, an INT4 Cosmos3-Edge reasoner on the same Orin, and a counted hidden-Markov filter that ties the recalibrated baseline from the detector score alone, with an analysis that locates the residual VLM gap in ego-relevance.

## II. Related Work

*a) Work-zone perception.:* Early systems used environmental models with sensor fusion [8] or frame classification [9]. ROADWork [5] provides 4,375 videos and 9.6K annotated images from 18 U.S. cities and shows large gains from in-domain fine-tuning (+32.2 AP, +36.7 SPICE). The closest system [3] deploys a ROADWork-trained YOLO detector with CLIP verification, EMA smoothing and hysteresis, and defines the traversal-state evaluation we adopt.

*b) VLMs and world models for driving.:* VLMs support planning [10], [11], description [12] and reasoning [13], and Cosmos world models plan at datacenter or automotive-SoC scale [14], [15]; we estimate one safety-relevant state under an embedded latency budget.

*c) Cascades.:* Detection [16] and LLM [17] cascades chain models of increasing cost and escalate on confidence; our stages share one model, and evidence semantics (transcribed text versus a binary answer) drive escalation.

*d) Temporal filtering and fair comparison.:* Our joint estimator is a discrete hidden Markov model [18] run as a Bayes forward filter [19]; we claim no novelty for it. That baselines deserve the proposed method's tuning budget is known [20]; we measure how much it changes conclusions for embedded temporal estimators.

## III. System

### A. Two embedded deployments

We deploy *two* models on the same Jetson AGX Orin 64 GB, which shares its SoC with the automotive DRIVE AGX Orin, adapted in two common ways: one by task fine-tuning, one by quantization and per-model calibration alone. Both run the same prompt channels, cascade structure, protocol and splits; only the model, its input size and its calibrated constants differ.

**Fine-tuned 2B.** A 2B-parameter VLM (Qwen3-VL architecture [21], initialized from Cosmos-Reason2-2B [22]), used for perception only, fine-tuned through a curriculum derived from ROADWork [5], whose imagery spans 18 U.S. cities under one signing standard (MUTCD [23]): work-zone VQA ($\sim$46K pairs), ego-state supervision, sign reading, a general-driving retention mix, and $\sim$10K work-zone-free BDD100K hard negatives. The last stage brought the existence gate from 95% to 0% "yes" on clean BDD100K scenes (on ROADWork OUTSIDE seconds, which include visible but irrelevant work, it is less selective, Table IV).

The language backbone is quantized to INT4 (W4A16 AWQ, group size 128) and the vision tower kept in FP16, both as on-device TensorRT engines [7] (1.38 + 0.83 GB). Decoding (10.4 ms/token) dominates cost, followed by vision encoding (44.6 ms at 720p, $\sim$20 ms at our 480px).

**Quantized 4B world model.** Cosmos3-Edge (C3E) [14] pairs an autoregressive reasoner with a trajectory-generating diffusion transformer; we run only the reasoner, and "world model" below refers to it. It is quantized to W4A16 INT4 (1.27 GB) with an FP16 visual tower (983 MB) on the same Orin; its frozen $26\times46$ patch grid fixes inputs at $736\times416$, and channel latencies are 162/182/488/556 ms (GATE/EGO/DESC/SIGN). It has seen no work-zone training data. Its engines required six patches to the embedded runtime, released upon acceptance.

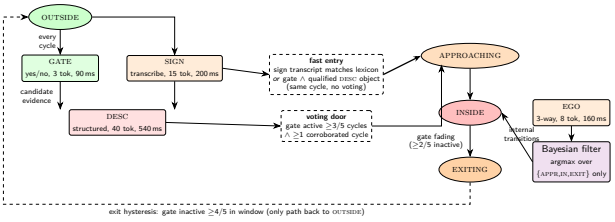



Fig. 2. Evidence-first cascade. In OUTSIDE, GATE and SIGN run every cycle and candidate evidence invokes DESC. Entry requires a sustained, corroborated vote or high-confidence same-cycle evidence. In active states, EGO drives a Bayesian filter; sustained gate clearance returns to OUTSIDE, while partial fading triggers EXITING. Constants are calibrated per model: $5/3/4/2$ for the 2B and $3/3/2/1$ for C3E, so the structure transfers but the numbers do not. Diagram values are the 2B's, with indicative latencies (measured values in Section V-D).

### B. Prompt channels and their failure modes

We operate four fixed-prompt channels with fixed token budgets (the first generated token is an answer-start token).

*a) Affirmation bias.:* The GATE asks a neutral existence question, but binary prompts cannot be specialized by rephrasing: path-relevance variants all answered "yes" on OUTSIDE probes and distance prompts "close" regardless of scene. Binary outputs are therefore weak evidence requiring a temporal vote, never sole triggers. At $\sim$10 ms per output token, a three-token answer costs $\sim$85 ms and a description $\sim$500 ms, so evidence quality, not only confidence, gates smoothing.

*b) Faithful transcription.:* The SIGN channel returns verbatim text such as "SHOULDER WORK" and correctly reports absence, while a binary rephrasing misses signs that transcription catches; we match transcripts against a work-zone lexicon.

*c) Structured corroboration.:* DESC often verbalizes disqualifying context ("barriers on right *sidewalk*", "work vehicle *parked*"), so corroboration requires a work-zone object without such a qualifier. Prompts, budgets and lexicons are released upon acceptance.

### C. Evidence-first cascade

The cascade (Fig. 2) allocates compute by state.

**Outside (patrol).** GATE and SIGN run every cycle (2B at 480px: p95 259 ms, 3.9 Hz), and either can invoke DESC. APPROACHING is entered by vote (gate active in $K_{\text{enter}}$ of the last $N$ cycles plus one corroboration) or by fast entry on a lexicon-matching sign or same-cycle gate+DESC agreement, which avoids a delay of up to $N$ cycles ($\sim$1.3 s for the 2B) for evidence such as "ROAD WORK AHEAD."

**Active states.** EGO updates a Bayesian filter whose argmax is restricted to the active states, so that one "outside" reply right after a sign-triggered entry cannot erase it. Only sustained gate clearance returns to OUTSIDE; partial fading triggers EXITING, which EGO never produced. DESC refreshes every

1.5 s. The 2B's thresholds ($N$=5, $K_{\text{enter}}$=3, $K_{\text{exit}}$=4, $K_{\text{fade}}$=2) were fixed on calibration data; C3E gets its own (Section VI). With EGO disabled, as selected for C3E, APPROACHING becomes INSIDE on the transition prior alone, 13 cycles after entry.

### D. Joint estimator over both evidence streams

Grafting the detector and the world model onto each other's temporal machinery does not work: the ported detector counts cycles (150-cycle maximum approach, 20 minimum frames outside) tuned for its native loop, and the cascade votes over a window tuned for the 2B's ∼260 ms cycle, while C3E answers only every ∼1.5 s. Either spine imports constants that are wrong for the combination.

The joint estimator therefore inherits no constant: it is a discrete hidden Markov model over the four states, run as a forward filter [18] with maximum-likelihood tables and no EMA, hysteresis threshold or cycle counter. Its observation combines the detector's frame score (six buckets), the world model's gate, sign-keyword and corroboration answers ordered by strength (six levels), and the age of that answer (three buckets): 108 cells. The emission is counted once per detector cycle on the calibration split with Laplace smoothing, jointly rather than factorized, because the interaction is the signal: the world model's APPROACHING is reliable when the detector corroborates it and mostly a false alarm when it does not (73% of those frames are OUTSIDE). The transition is a per-second rate matrix, counted as ground-truth transitions over dwell time and scaled by each step's $dt$, so it does not depend on the loop rate. The log-likelihood is scaled by $dt/\tau$ ($\tau = 0.5$ s): a world-model answer persists across ∼30 detector frames, and counting it once per frame drove an early version to 135 false alarms per hour. Bucket edges and $\tau$ were set on calibration data (Section VII).

The detector runs at its native rate and is never blocked; the world model is re-polled as soon as it answers, and its last answer is held with its age. We record raw evidence with measured clocks once per video and replay the filter on it, so ages and frame drops in every result are measured ones; Section VII confirms the replay against live concurrent execution.

## IV. Evaluation Protocol

*a) Benchmark.:* We use the detector baseline's 520-video ROADWork traversal benchmark [3]: 4.31 h of 1920×1080 driving from 25 drives (30 fps except seven 120 fps videos), with 484K interval-labeled frames. A state-stratified 60/40 split assigns all design choices to calibration (312 videos, 2.59 h) and results to validation (208 videos, 192K frames, 1.73 h, 13 pure-negative). All 25 drives occur in both partitions, so Section VIII repeats every replay-fitted comparison with drive-disjoint folds. ROADWork samples its annotated images from the same drives as its videos, so the 2B's fine-tuning images, like the baseline detector's training images, may come from benchmark drives; C3E is zero-shot. Every calibration effect we report compares a system with itself, perception fixed, and is not inflated by this overlap.

*b) Latency-honest camera simulation.:* After each measured inference cycle of duration $\Delta$ on the target hardware, the replay advances by $\Delta$ and drops intervening frames as a live camera would, so slower systems see sparser frames and pay latency inside the accuracy result. Predictions are held between cycles and scored per frame.

*c) Metrics.:* We adopt the baseline's suite [3]: frame accuracy and macro-F1; per-state IoU; event precision/recall (an interval is correct if it overlaps one of the same state); entry offsets (Entry MAE over all states, IN entry med. for INSIDE); and false alarms per hour (predicted active intervals overlapping no annotated one), plus a lead metric for timing (Section V-A). Per-hour rates count frames at 30 fps (validation time overstated by 3%, for all systems alike). Intervals are paired 95% cluster-bootstrap intervals (10,000 replications) that resample drives.

TABLE I

VALIDATION, 208 VIDEOS, ONE PROTOCOL AND DEVICE (%; TIMING IN S). DET.+CLIP: HAND-TUNED [3] AND RECALIBRATED (SECTION VI); DET. *cnt.*: DETECTOR SCORE IN THE COUNTED FILTER; 2B: GREEDY, NO POST-FILTER (RECALIBRATED: 0.459); JOINT: DETECTOR+C3E IN THE COUNTED FILTER. LEAD: INSIDE ONSETS PRECEDED BY ≥2 S OF ACTIVE STATE. F1@20: AT ≤20 FALSE ALARMS/H (SECTION VIII).

| Metric | Det.+CLIP hand | Det.+CLIP cal. | Det. cnt. | 2B | C3E | Joint |
|---|---|---|---|---|---|---|
| Frame acc. | 62.9% | 64.8% | **68.2%** | 62.0% | 65.5% | 66.1% |
| Macro F1 | 0.470 | **0.551** | 0.524 | 0.453 | 0.515 | 0.546 |
| IoU OUT | 0.617 | 0.627 | **0.652** | 0.581 | 0.615 | 0.641 |
| IoU APPR | 0.124 | **0.296** | 0.219 | 0.152 | 0.278 | 0.246 |
| IoU IN | 0.544 | 0.552 | 0.573 | 0.532 | 0.574 | **0.578** |
| IoU EXIT | 0.106 | 0.153 | 0.124 | 0.062 | 0.071 | **0.160** |
| IN ev. recall | 95.7 | 77.8 | 88.6 | **97.3** | 87.0 | 85.9 |
| IN ev. prec. | 79.9 | **92.5** | 75.9 | 56.8 | 71.5 | 66.8 |
| False alarms/h | 30.9 | **20.2** | 22.5 | 56.2 | 64.6 | 25.9 |
| Entry MAE | 2.52 | 2.31 | 2.44 | 2.52 | **2.15** | 2.22 |
| IN entry med. | 3.47 | **2.10** | 2.73 | 2.67 | 2.50 | 2.27 |
| Lead ≥2 s | 79.4 | 78.8 | 73.1 | 76.2 | **86.2** | 66.9 |
| F1@20 | 0.410 | 0.529 | 0.520 | 0.409 | 0.487 | **0.537** |

## V. The VLM Cascade Against the Published Baseline

### A. Comparison with the deployed detector system

Table I places every system on the same videos, device and protocol; this section compares the 2B with the detector pipeline as published. With greedy decoding, our default (Section V-C), the 2B is statistically tied with it on frame accuracy and macro-F1 ($-0.018$, $[-0.041, +0.003]$) and matches its INSIDE event recall, but trails on event precision (56.8% vs. 79.9%) and false alarms (56.2 vs. 30.9 per hour): a competitive detector-free result, not a replacement. A causal 1 s INSIDE debounce raises precision to 76.0% (macro-F1 0.478) without retraining. Entry MAE is equal (2.52 s), and the 2B's INSIDE entry median is 0.80 s better.

TABLE II
SINGLE-TOGGLE ABLATIONS OF THE 2B CASCADE (VALIDATION, 208 VIDEOS; –: NO INSIDE EVENT PREDICTED). LEAD: INSIDE ONSETS PRECEDED BY ≥2 S OF ACTIVE STATE.

| Variant | Acc. | F1 | IN prec. | FA/h | Lead |
|---|---|---|---|---|---|
| Cascade, sampled | 62.5 | 0.464 | 54.1 | 60.1 | 79.4 |
| greedy decoding | 62.0 | 0.453 | 56.8 | 56.2 | 76.2 |
| binary SIGN | 52.5 | 0.407 | 46.1 | 305.8 | 86.2 |
| no fast entry | 63.1 | 0.462 | 54.3 | 34.3 | 69.4 |
| no qualifier filter | 60.3 | 0.447 | 53.9 | 69.7 | 81.2 |
| no SIGN channel | 63.2 | 0.456 | 53.4 | 47.8 | 82.5 |
| unrestricted filter | 61.4 | 0.445 | 53.1 | 69.1 | 73.8 |
| base checkpoint | 48.8 | 0.289 | – | 29.2 | – |

Both pipelines handle established INSIDE intervals well and EXITING poorly. The 2B's APPROACHING IoU is higher (0.152 vs. 0.124), as text and distant evidence appear before detector counts stabilize, a gap the recalibrated detector closes (Section VI).

### B. Operational false-alarm characterization

On validation the 2B raises 56.2 false active-state events per hour, against 30.9 for the detector pipeline; most are the ego-relevance errors of Section IX, not clean-scene noise. Of 156 matched INSIDE entries, 80.8% fire early (median 2.9 s), which an advisory prefers. Section VIII compares all systems at matched false alarms.

### C. Ablations and design validation

Table II toggles one component at a time from the sampled cascade (first row); five of seven change macro-F1 by at most 0.02. The sign channel must transcribe: an equal-cost binary question cuts accuracy from 62.5% to 52.5% and raises false alarms fivefold. Greedy decoding costs 0.5 accuracy points but improves precision and alarms, so we use it as the reproducible default. The qualifier filter adds 2.2 points and removes 9.6 alarms per hour. Fast entry buys anticipation, 10 more points of onsets warned ≥2 s ahead, for 26 more alarms per hour. The base checkpoint collapses under the 2B's thresholds (Section VI).

### D. Operational cost

At 480px, p50/p95 latency is 83/90 ms (GATE), 136/149 (EGO), 155/169 (SIGN) and 468/508 ms (DESC); patrol and active cycles run at p95 259 ms (3.9 Hz) and 239 ms (4.2 Hz), the heaviest path at 768 ms. "Real time" means keeping pace with a live camera (259 ms is 7.8 m at 30 m/s), not an alert deadline. Above idle, the VLM draws 28.6 W and 3–26 J per cycle, the detector 22.4 W and 0.5 J at 24 ms.

## VI. CALIBRATING EVERY SYSTEM

Section V ran each VLM with thresholds fitted to the 2B and the detector pipeline with its published constants. Here every system is recalibrated by the same replay procedure, and the ranking reverses (Fig. 1).

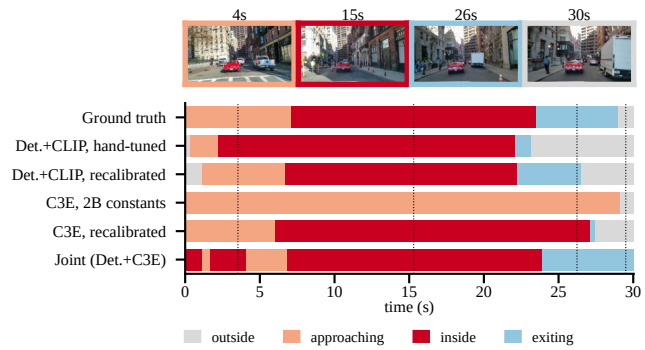



Fig. 3. Ground truth and five systems on one validation video, chosen to illustrate the effect; thumbnails mark the middle of each annotated state. With the 2B's constants, C3E never leaves APPROACHING; recalibrated, it tracks the zone. The hand-tuned detector pipeline enters INSIDE 5 s early and cuts EXITING short; recalibrated, it follows all four states.

*a) Temporal hyper-parameters do not transfer.:* With the 2B's constants, the zero-shot C3E almost never reaches INSIDE (IoU 0.007, recall 3.8%) and looks useless. Recalibrating the same four integers and two switches lifts it to macro-F1 0.515 and INSIDE IoU 0.574, with model, prompts and videos unchanged (Fig. 3). The search covers $N \in \{3, 5, 7, 9\}$, $K_* \in 1..N$ and the EGO/SIGN switches (4,896 configurations, later extended to $N \in \{1, 2, 4\}$ without changing the optimum), selected on calibration macro-F1 over the 100 videos with recorded VLM evidence. The optimum ($N$=3, $K$=3/2/1) is shorter and less hysteretic and disables EGO (Section IX). Because each channel is recorded once and the state machine replayed, a sweep costs CPU-seconds instead of GPU-hours.

The same search moves the fine-tuned 2B by only +0.006 (0.453→0.459): it rescues a model whose constants were wrong rather than rewarding flexibility.

*b) The baseline too.:* The detector pipeline deserves the same treatment. We record every component on every cycle of all 520 videos (detector counts before and after CLIP verification, global CLIP and colour-context scores) and replay the complete pipeline; its hand-tuned configuration reproduces the live run (0.473 vs. 0.470). Searching its thresholds, EMA rate and duration limits on the 312 calibration videos (three grids of 4,800 configurations) raises validation macro-F1 from 0.470 to 0.551 ($+0.081$, $[+0.043, +0.111]$), with 92.5% INSIDE precision at 20.2 false alarms per hour: the strongest system we measure. On the detector score alone, without CLIP, the same search gives 0.435→0.530.

The procedure also isolates what fine-tuning buys. Recalibrated on its own evidence, the base checkpoint, identical except for the work-zone curriculum, reaches 0.383 on the search videos against 0.496 for the fine-tuned model: fine-tuning is worth $+0.113$ macro-F1, not the collapse (INSIDE IoU 0.012) the base model shows under the fine-tuned model's thresholds.

Every choice is made on the 312 calibration videos and none on validation: prompts and lexicons on the full partition, VLM thresholds on the 100-video subset with recorded VLM

evidence, detector thresholds and counted tables on all 312. The budgets therefore favour the detector; given the VLM's budget, one grid on the same 100 videos, the recalibrated pipeline still reaches 0.522, $+0.052$ over its hand-tuned version ($[+0.038, +0.070]$) and level with C3E.

*c) No configuration dominates.:* Calibrated, the world model and the detector pipeline tie on macro-F1 ($-0.036$, $[-0.080, +0.005]$) and fail differently: C3E leads INSIDE IoU and entry MAE, the detector pipeline precision and nuisance (20.2 vs. 64.6 false alarms/h). Section VII combines them.

## VII. Combining the Detector and the World Model

The joint estimator reaches 0.546 at 25.9 false alarms per hour. It beats the hand-tuned pipeline ($+0.076$, $[+0.038, +0.104]$) and the calibrated world model ($+0.031$, $[+0.006, +0.058]$, with 38.7 fewer alarms per hour), and ties the recalibrated pipeline ($-0.005$, $[-0.046, +0.033]$), trading INSIDE precision (66.8% vs. 92.5%) for recall (85.9% vs. 77.8%).

Its tables are counted on all 312 calibration videos (5-fold cross-validation: 0.535, s.d. 0.019), and its design choices behave predictably: $\tau$ trades accuracy for nuisance (0.538 at 44.7/h for $\tau$=0.25, 0.504 at 22.0/h for $\tau$=1; we keep 0.5), 20 of the 108 cells carry 97% of the observations, and 3 or 10 detector buckets give 0.538 and 0.545.

*a) Concurrent execution.:* The recorded runs behind every result had a 44 ms detector cycle (p50; p95 59 ms) and a world-model answer every 1.54 s. Running both models truly concurrently on the Orin, in wall-clock time from a decoded camera stream, on all 208 validation videos, raises the detector cycle from 24 ms alone to 53 ms (p95 58 ms, 19 Hz) with an answer every 1.44 s. Live, the joint estimator scores 0.548 against 0.546 in replay ($+0.002$, $[-0.011, +0.020]$; false alarms $+0.6$/h, $[-4.7, +5.6]$) and the detector-only filter 0.522 against 0.524: the replay is faithful under real GPU contention. The Orin draws 39.0 W median (46.9 W peak) with both models, 33.5 W with the detector alone and 11.1 W idle.

*b) Where the gain comes from.:* The detector carries the filter: without it macro-F1 drops by 0.086 and EXITING IoU from 0.160 to 0.017, while removing the world model costs 0.022 and lowers alarms (25.9 to 22.5 per hour). The detector score alone in the counted filter scores 0.524 at 22.5 false alarms per hour, tied with the recalibrated detector+CLIP pipeline on both ($-0.026$, $[-0.062, +0.009]$; $+2.2$/h, $[-8.1, +11.1]$) without CLIP or a threshold search, and 11.2 alarms per hour below the state machine recalibrated on the same score ($[-24.6, -2.1]$): the most efficient option here. The world model pays where the detector is weak: on the 10 validation onsets preceded by a transcribed work-zone sign, the joint estimator and C3E warn $\geq$2 s ahead on all ten, the recalibrated detector+CLIP pipeline on eight.

*c) A learned alternative.:* A strictly causal three-stage TCN [24] trained on the same evidence ties the best macro-F1 (0.550) but at 71.9 false alarms per hour.

TABLE III
DRIVE-DISJOINT 5-FOLD REPLAY, ALL 520 VIDEOS: EVERY SYSTEM FITTED WITHOUT THE TEST DRIVES. F1@20: MACRO-F1 AT $\leq$20 FALSE ALARMS PER HOUR UNDER A SHARED PERSISTENCE FILTER.

| System | F1 | IoU IN | IN prec. | FA/h | F1@20 |
|---|---|---|---|---|---|
| Det.+CLIP, hand-tuned | 0.481 | 0.570 | 79.8 | 26.1 | 0.417 |
| Det.+CLIP, recalibrated | 0.535 | 0.560 | **90.7** | **18.7** | **0.535** |
| Det. score, hand-tuned | 0.434 | 0.546 | 73.3 | 47.0 | 0.410 |
| Det. score, recalibrated | 0.523 | 0.543 | 86.7 | 43.9 | 0.420 |
| Det., counted filter | 0.524 | 0.590 | 73.5 | 22.5 | 0.520 |
| Det.+C3E, counted filter | 0.537 | 0.591 | 70.8 | 28.7 | 0.523 |

## VIII. Do the Conclusions Hold?

*a) Unseen drives.:* Splitting the 25 drives into five folds of about 104 videos, we refit the detector state machines (full three-grid search) and the counted filters on four folds and replay them on the fifth (standalone VLM evidence covers only 100 videos), so no test video shares a drive with its fitting data (Table III). The conclusions hold: recalibration raises the detector+CLIP pipeline by 0.054 macro-F1 ($[+0.026, +0.078]$) and the detector-score state machine by 0.089 ($[+0.058, +0.114]$), and both counted filters tie the recalibrated pipeline ($-0.010$, $[-0.036, +0.017]$; $+0.002$). On the validation videos, drive-disjoint fitting moves each system's macro-F1 by at most 0.010 from Table I.

*b) Matched nuisance.:* Systems operate between 18 and 65 false alarms per hour. Passing each through the same causal persistence filter (an active interval is shown only after lasting $d$ seconds) until false alarms fall to 20/h (last column of Tables I and III), the calibrated systems lead: on drive-disjoint folds the recalibrated detector+CLIP pipeline keeps 0.535 at $d$=0, the joint estimator 0.523 and the counted detector filter 0.520 at $d \leq 0.75$ s, against 0.42 for the hand-tuned pipeline. Matching nuisance also reorders anticipation: at $\leq$30 false alarms per hour, C3E warns $\geq$2 s before 58% of INSIDE onsets and the 2B before 54%, against 73% for the counted detector filter and 79% for the recalibrated pipeline.

*c) Another platform.:* We reran the hand-tuned detector+CLIP, both 2B decodings, C3E and the joint estimator on a DRIVE AGX Thor with Orin-fitted constants and tables. Model answers agree with the Orin's on 96–100% of identical frames, but every stage is faster (detector 16 vs. 24 ms; C3E's SIGN 157 vs. 556 ms), so cycle-counted constants span less time. The hand-tuned detector pipeline loses 0.031 macro-F1 ($[-0.037, -0.023]$) and C3E 0.042 ($[-0.055, -0.026]$), with 21 and 41 more false alarms per hour; the decode-bound 2B loses little macro-F1 ($-0.007$), though its alarms also rise by 21 per hour. The joint estimator, whose rates are per second and whose evidence is weighted by $dt$, keeps its macro-F1 without recalibration (0.549 vs. 0.546, $[-0.018, +0.028]$). New hardware is the same hidden variable as a new model.

## IX. Failure Analysis: the Ego-Relevance Gap

Nearly all residual false positives are correct descriptions of objects irrelevant to the vehicle's path (a parked work vehicle 60 m away, sidewalk work behind barriers, detour signs for

TABLE IV
GATE "YES" RATE BY TRUE STATE, INSIDE-VS-OUTSIDE BALANCED ACCURACY OF THE BINARY ANSWER (BA), AND DESC CORROBORATION ON OUTSIDE; SAME 100 CALIBRATION VIDEOS, 1 HZ.

| | GATE "yes" when truth is | | | | | |
|---|---|---|---|---|---|---|
| Checkpoint | OUT | APPR | IN | EXIT | BA | DESC |
| 2B base | 3.8% | 53.1% | 71.8% | 9.8% | **0.84** | 31.3% |
| 2B fine-tuned | 42.4% | 85.3% | 92.2% | 51.2% | 0.75 | **10.7%** |
| C3E zero-shot | 29.4% | 93.3% | 94.0% | 34.4% | 0.82 | 35.7% |

another street); the qualifier filter catches only what the model verbalizes, and binary or distance prompts do not recover the judgment.

*a) Response bias.:* Recording the three checkpoints on the same videos separates how often a model says "yes" from how well it tracks the state (Table IV). The base 2B rarely says yes outside a zone and separates INSIDE from OUTSIDE best (balanced accuracy 0.84); fine-tuning moves the gate to 42.4% and 0.75, often correct perception without ego-relevance, as ROADWork's OUTSIDE includes visible but irrelevant work. The fine-tuned model's advantage comes from its description, which corroborates on 10.7% of OUTSIDE seconds against 31.3% for the base model.

C3E, never fine-tuned, transcribes signs correctly and emits plausible boxes, yet labels most verified INSIDE frames "off-road/sidewalk" in ego-lane probes.

Perceiving objects and judging whether they claim the ego path are distinct capabilities; no prompt we tried bridged them in either model, across two scales and two adaptation strategies. The gap also explains weak EXITING IoU for language-only systems: receding objects remain visible. The joint estimator reaches 0.160 on EXITING only through the detector, whose score decays as objects recede (0.017 without it). ROADWork's 129K path annotations [5], unused in our fine-tuning, could supply trajectory-grounded ego-relevance labels.

## X. LIMITATIONS

Our claims are bounded in four ways. *(i) Data:* all footage is ROADWork daytime driving; drive-disjoint folds cover the temporal models, while the 2B's absolute accuracy may benefit from drive overlap with its fine-tuning images (Section IV). *(ii) Baseline:* comparisons use our port of the detector pipeline (0.470 on our split; 0.600 published on the original one). *(iii) Search:* configurations are the best found on finite grids; conclusions hold with matched budgets. *(iv) Scope:* replay is open-loop; we claim computational feasibility, not safety readiness.

## XI. CONCLUSION

Comparing an embedded VLM with a detector pipeline as published measures their temporal constants as much as their perception. Inherited constants move INSIDE IoU by 0.567; calibrating the baseline lifts its macro-F1 from 0.470 to 0.551, where no VLM-based system we built beats it, and a counted hidden-Markov filter on the detector score ties it without a search. This holds on unseen drives, at matched nuisance and budgets and in live concurrent execution, and on a DRIVE AGX Thor cycle-counted constants break while the counted filter holds. Embedded comparisons should recalibrate every system, the baseline included, on the target hardware, and compare at matched nuisance. Upon acceptance we release the replay harness, prompts, metrics and baseline port.

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